



24TH MEXICAN INTERNATIONAL
CONFERENCE ON ARTIFICIAL
INTELLIGENCE | **CIMAT & UG**

Water Quality Classification via Cost-Efficient Machine-Learning: A Case Study in Nuevo León

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**Tecnológico
de Monterrey**

Overuse, climate change and diffuse pollution



If you cannot measure it, you cannot improve it

- Lord Kelvin

Water Quality Index

Dinius

Classification	Value
Excelent (E)	90-100
Acceptable (A)	70-90
Slightly contaminated (LC)	50-70
Contaminated	30-50
Heavily contaminated	20-30
Extremely contaminated	0-20

$$WQI = \exp \left(\sum_{i=1}^n W_i \log Q_i \right), \quad Q_i \in (0, 100].$$

- temperature
- pH
- hardness
- coliforms

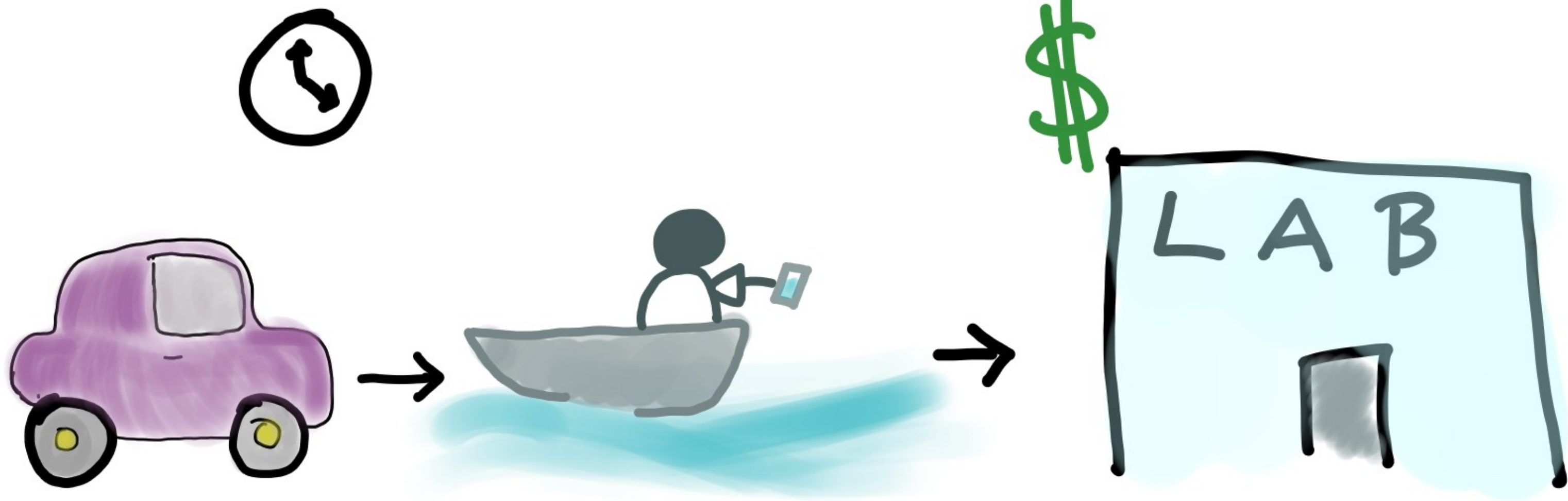
The problem

Measuring water is...

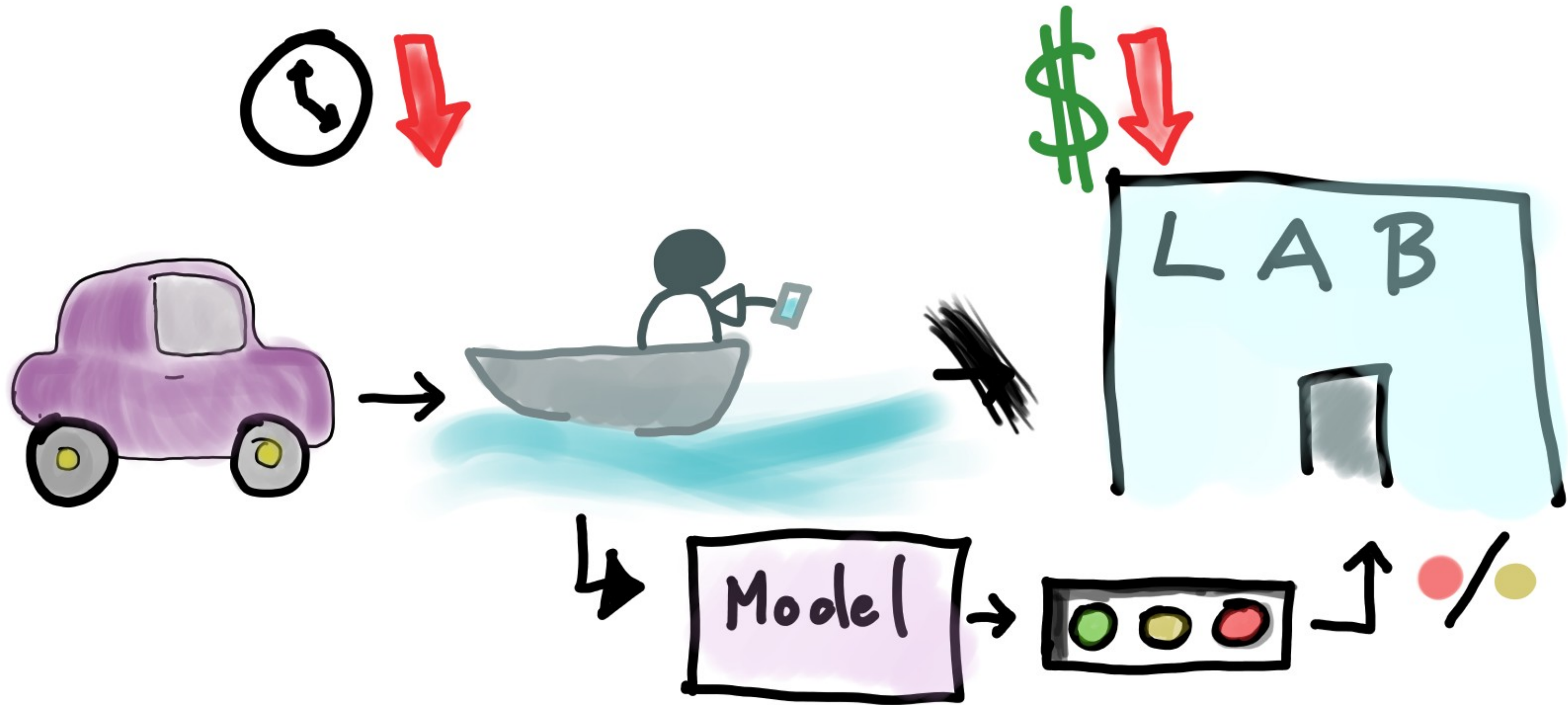
...expensive

...time consuming

...challenging



The proposal



Methodology

REMANECA DATASET

163 stations

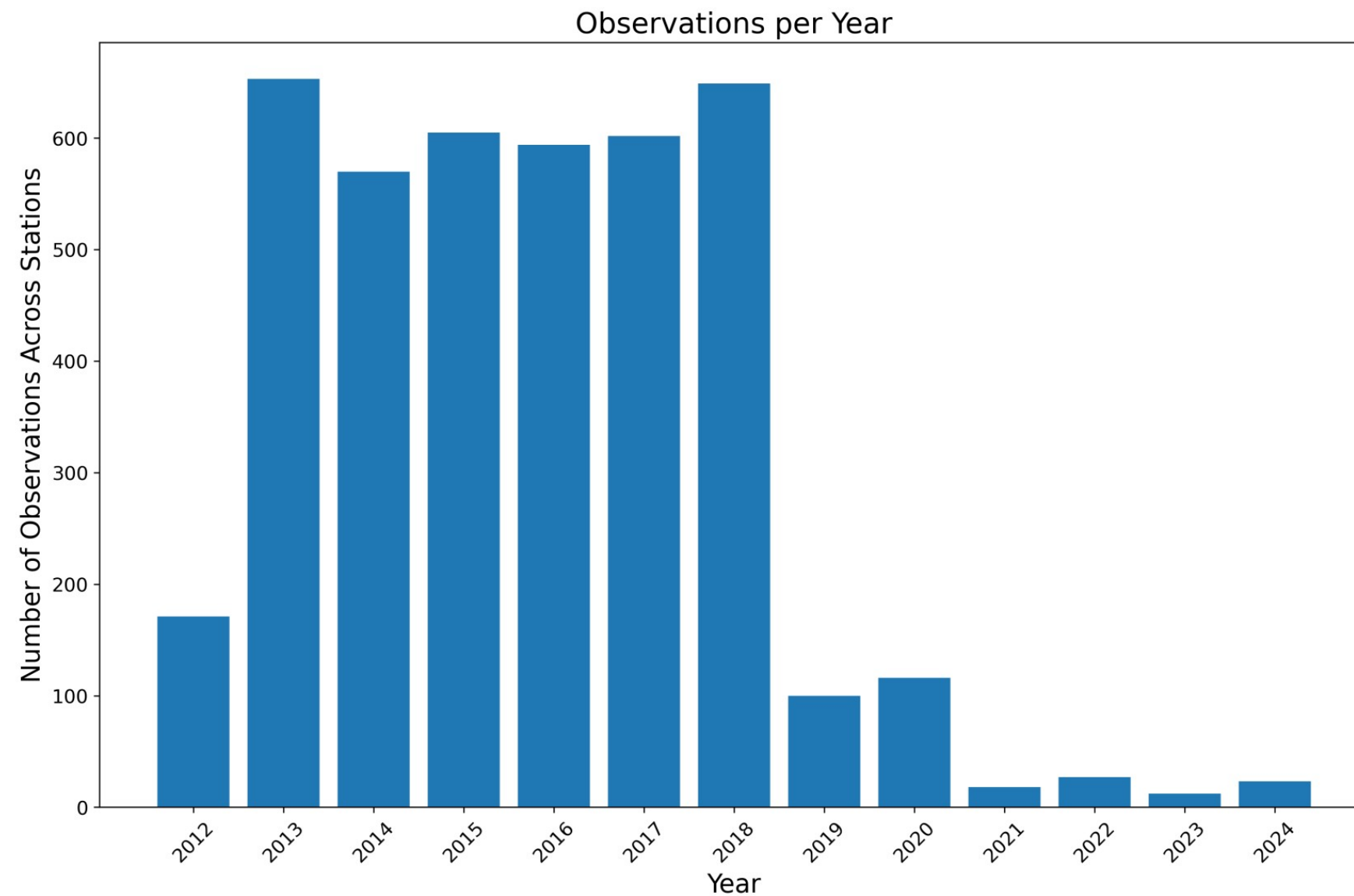
- Nuevo León, México
- 4,140 samples (2012 - 2024)
- 192 features
- Irregular sampling



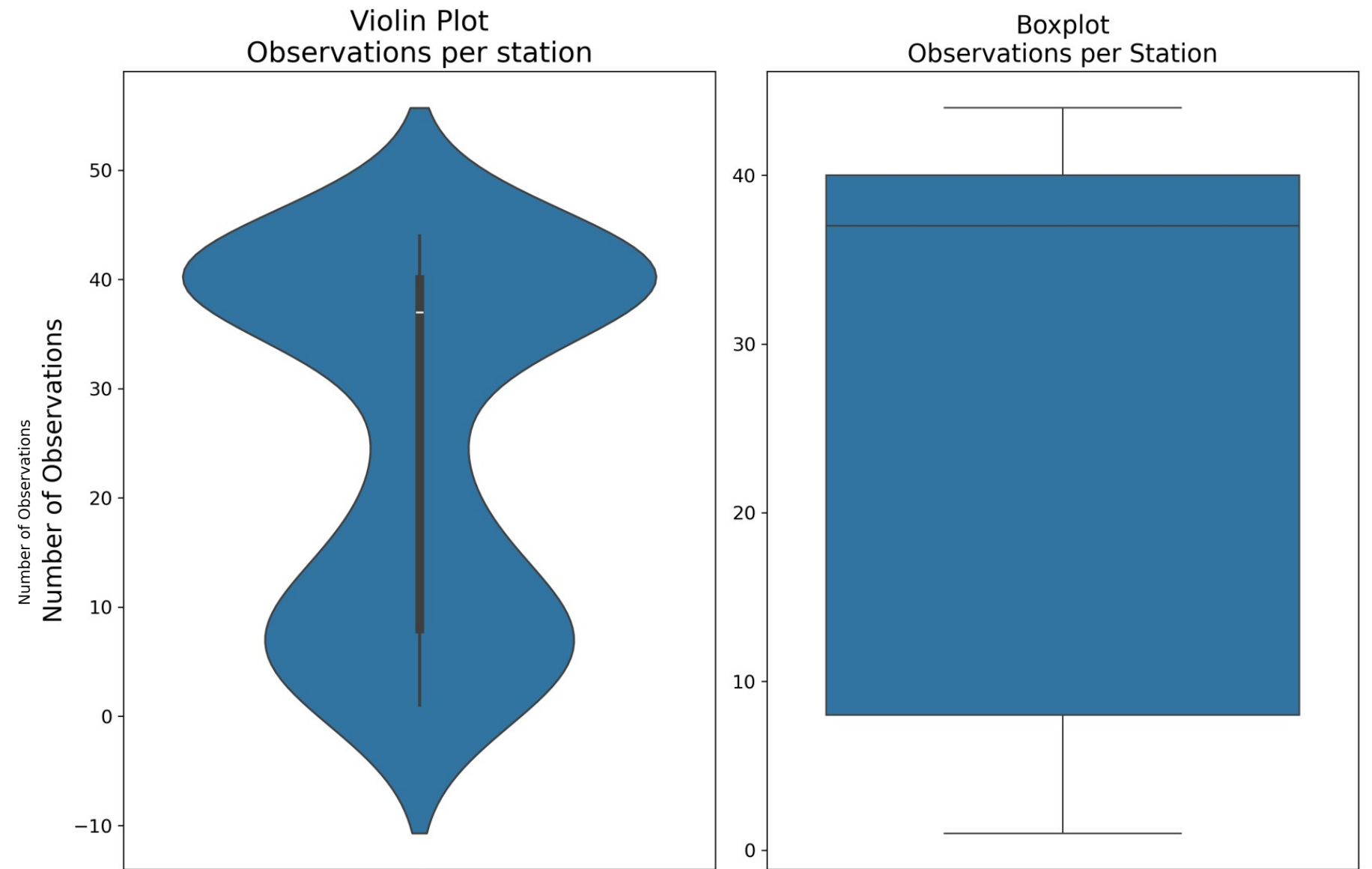
Github: [kaledcorona/water-quality-classification-paper](https://github.com/kaledcorona/water-quality-classification-paper)

Dataset description: 2012-2024

A)

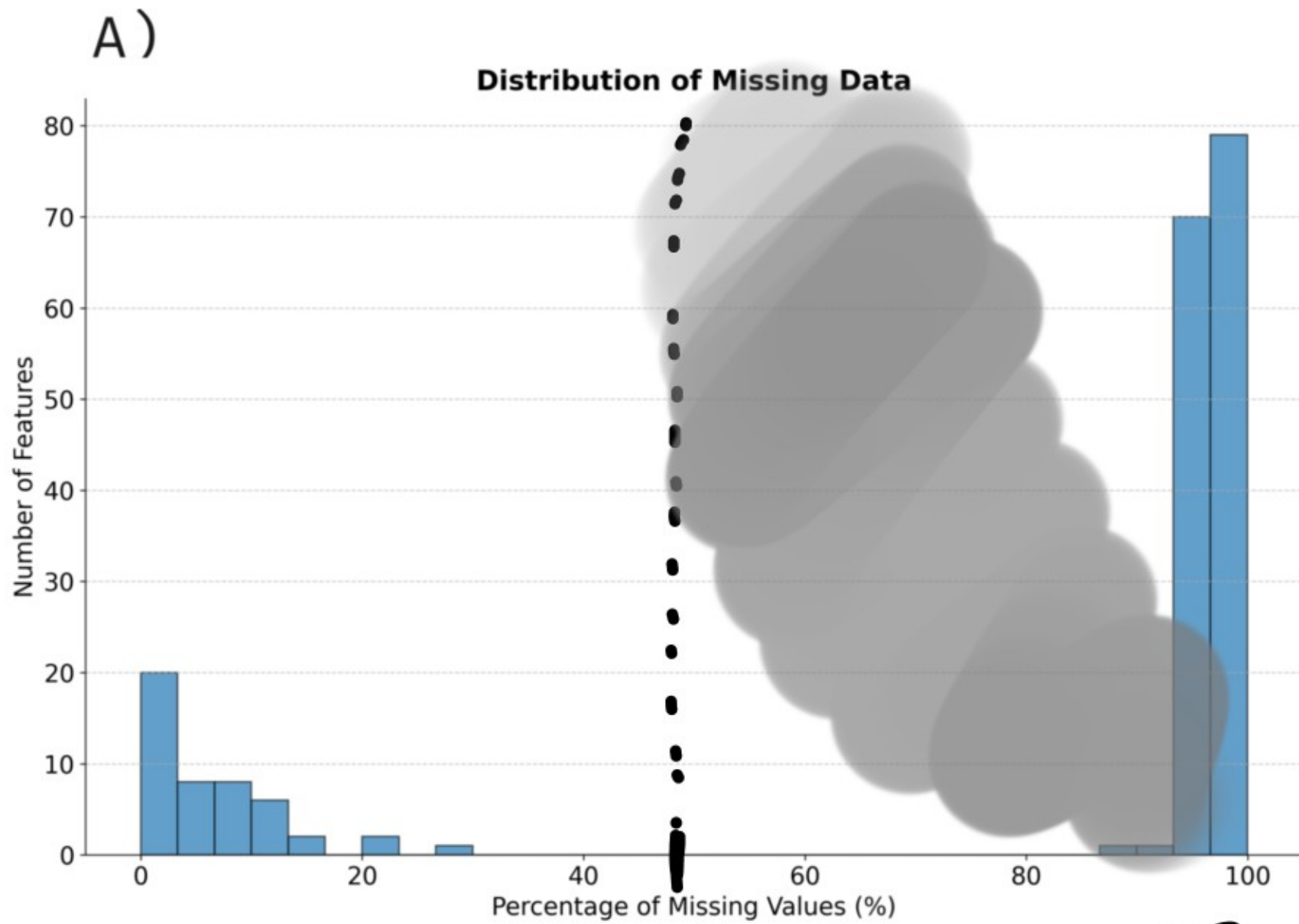


B)



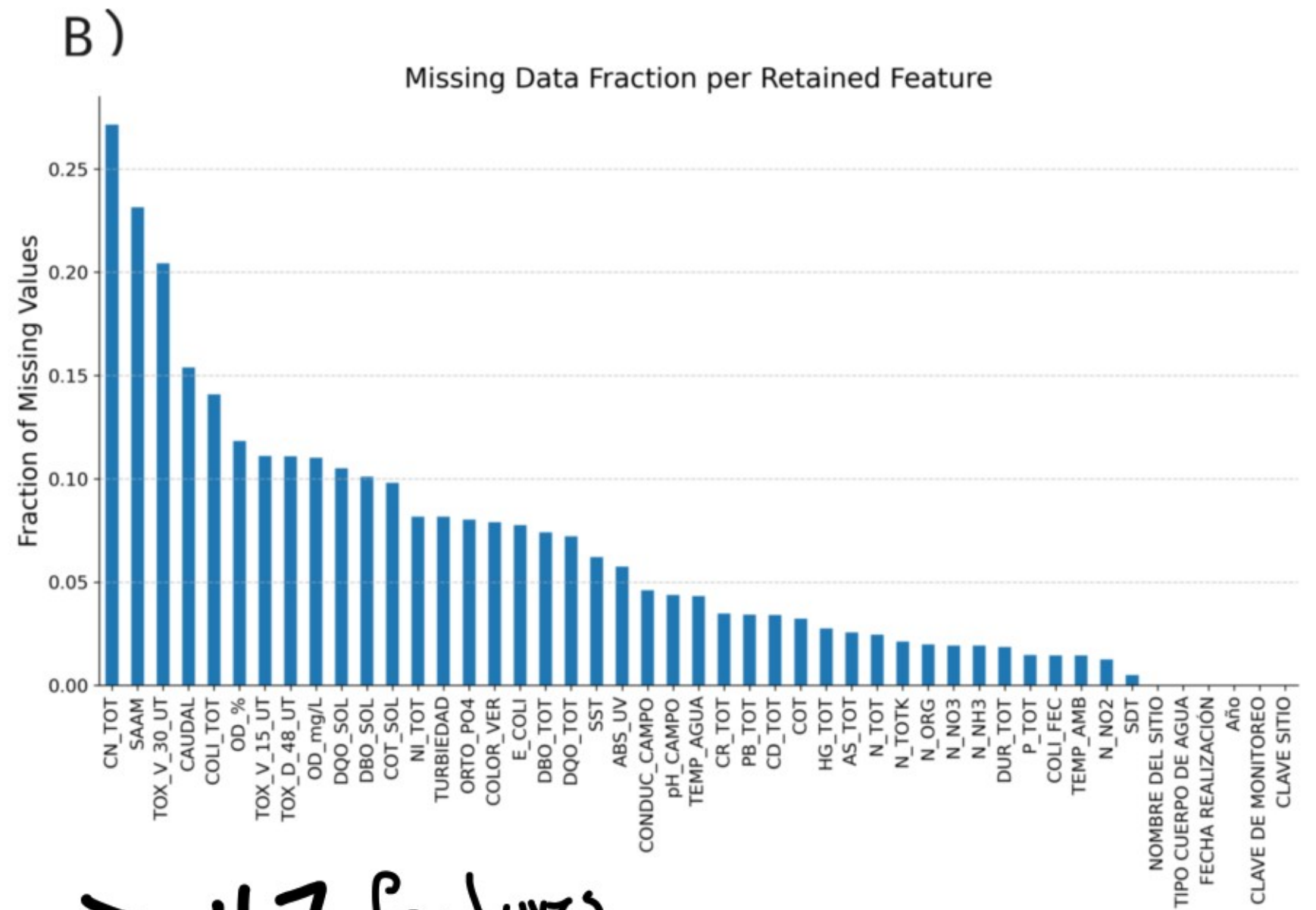
- Average of 25 samples per station (min 1, max 44)
- Irregular sample frequency
- Remove 50 stations due only 1 sample

Data preprocessing

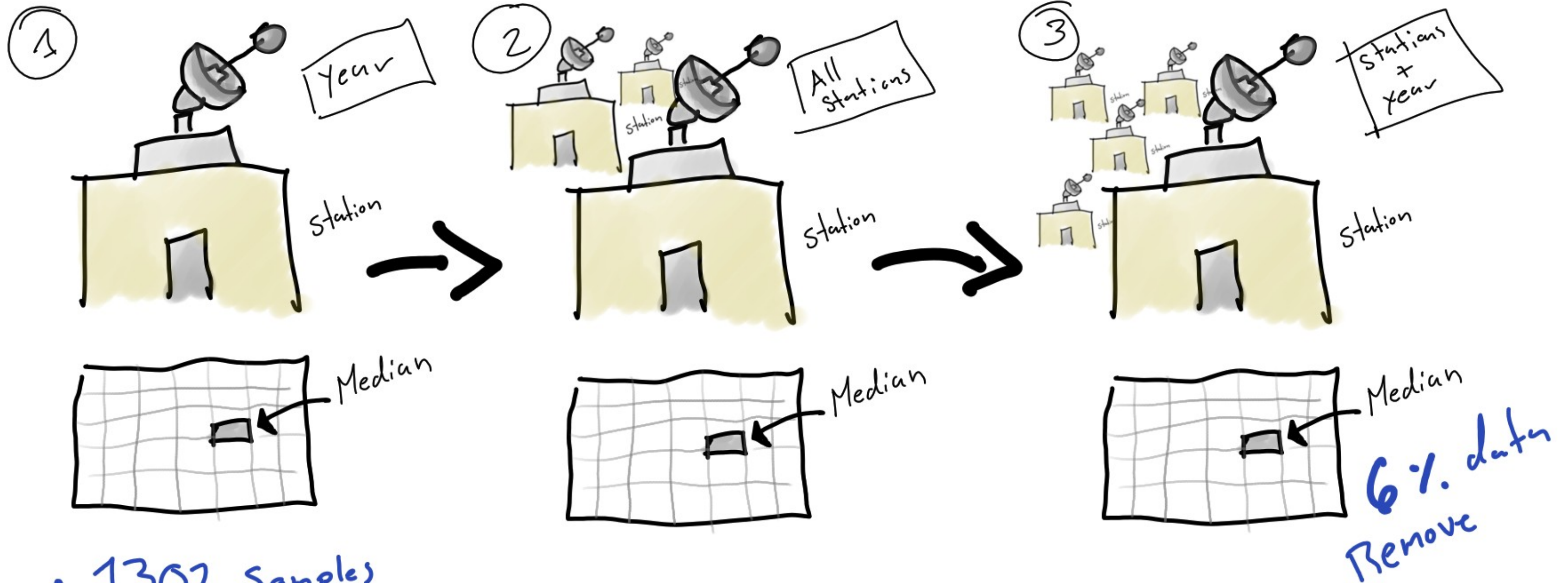


50%.

192 → 47 features



Data imputation



- 1302 samples
- 47 features
- Time: 2012-2020

Water Quality Index

①



②

Weights

Temperature $\rightarrow 0.21$

PH $\rightarrow 0.79$

③

$$WQI = \exp \left(\sum_{i=1}^n W_i \log Q_i \right), \quad Q_i \in (0, 100].$$

④

Categories

- Excellent: 90-100
-
- Heavily contaminated: 20-30

⑤

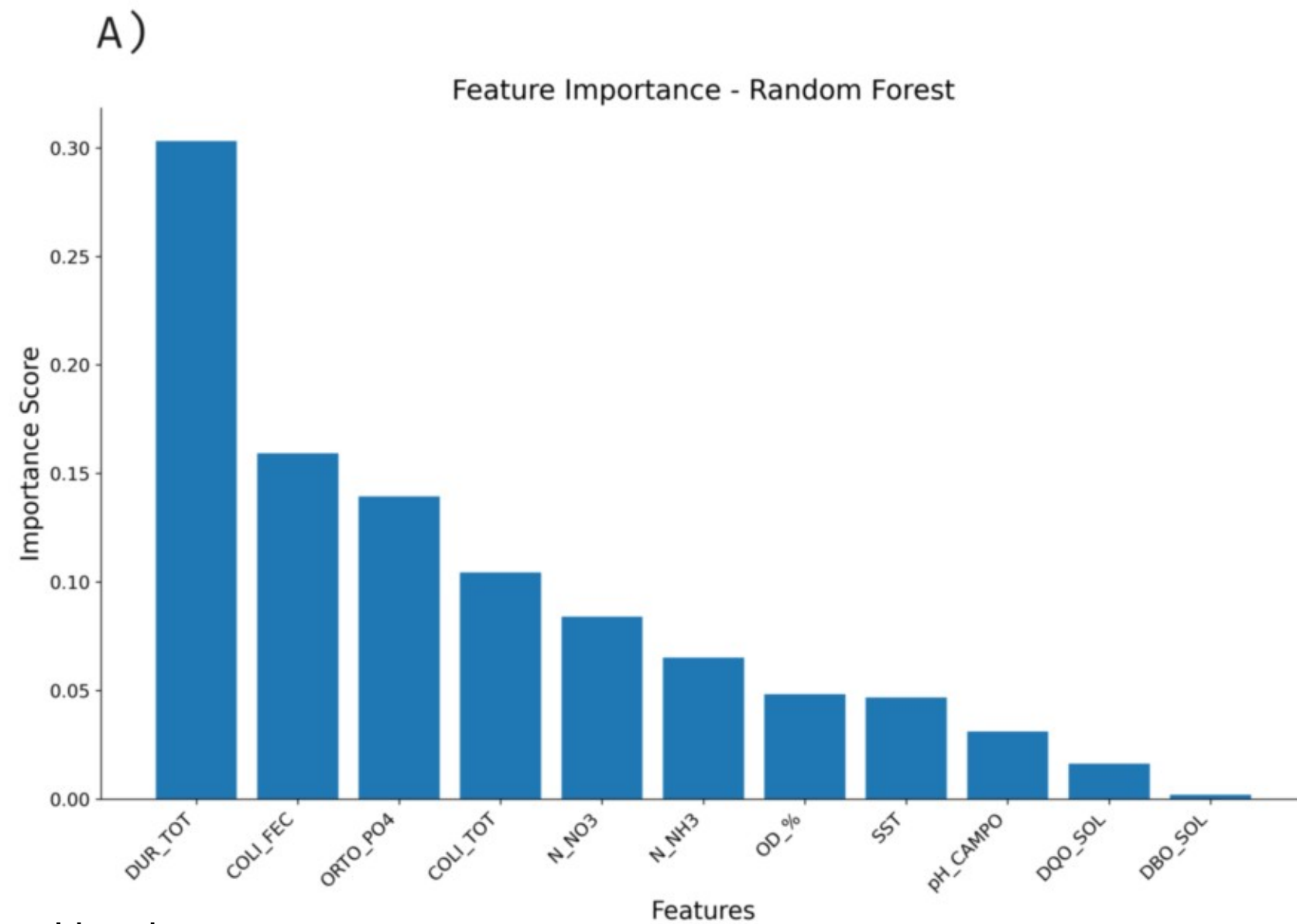
Category $\rightarrow$ ~~N~~ ordinal

10-fold Cross-validated performance (80/20)

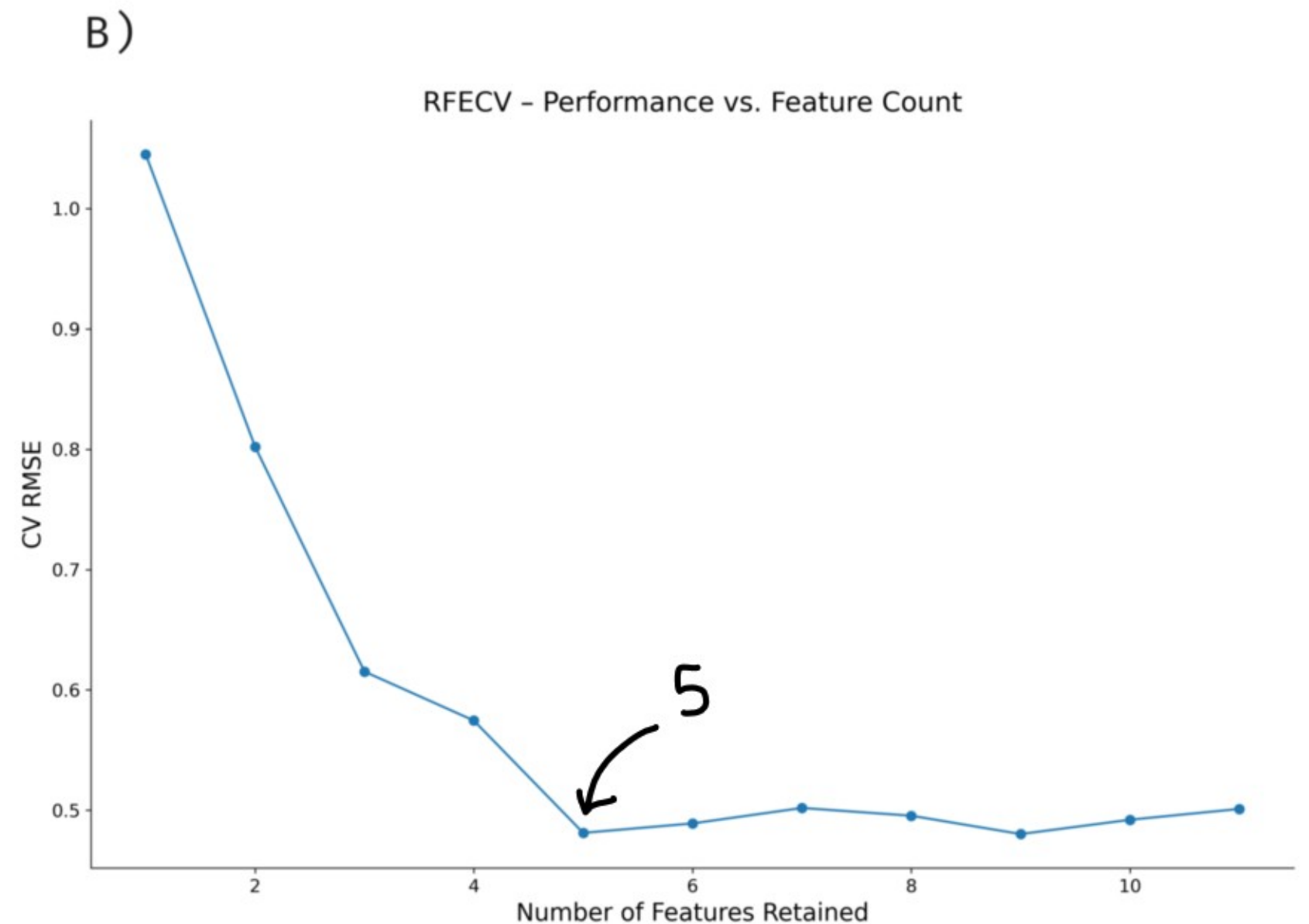
	DT	KNN	NB	RF	SVM
accuracy	0.898 $\pm$ 0.020	0.822 $\pm$ 0.029	0.793 $\pm$ 0.039	0.921 $\pm$ 0.023	0.866 $\pm$ 0.019
precision macro	0.829 $\pm$ 0.049	0.714 $\pm$ 0.077	0.682 $\pm$ 0.060	0.926 $\pm$ 0.037	0.653 $\pm$ 0.037
recall macro	0.830 $\pm$ 0.036	0.604 $\pm$ 0.059	0.815 $\pm$ 0.063	0.785 $\pm$ 0.073	0.634 $\pm$ 0.045
f1 macro	0.823 $\pm$ 0.031	0.641 $\pm$ 0.067	0.719 $\pm$ 0.060	0.831 $\pm$ 0.061	0.635 $\pm$ 0.043
precision weighted	0.904 $\pm$ 0.018	0.809 $\pm$ 0.039	0.859 $\pm$ 0.032	0.923 $\pm$ 0.022	0.810 $\pm$ 0.020
recall weighted	0.898 $\pm$ 0.020	0.822 $\pm$ 0.029	0.793 $\pm$ 0.039	0.921 $\pm$ 0.023	0.866 $\pm$ 0.019
f1 weighted	0.898 $\pm$ 0.019	0.808 $\pm$ 0.034	0.808 $\pm$ 0.036	0.912 $\pm$ 0.028	0.833 $\pm$ 0.022

DT: Decision Tree; k-NN: K-Nearest Neighbors; NB: Naïve Bayes; RF: Random Forest; SVM: Support Vector Machine.

Recursive feature elimination



- 1) Total hardness
- 2) Fecal coliforms
- 3) Orthophosphate
- 4) Total coliforms
- 5) Nitrate
- 6) Ammonia



Experimenting with 5 selected features

Table 3. Cross-validated performance with five-feature models (mean $\pm$ SD across folds).

	DT	KNN	NB	RF	SVM
accuracy	0.783 $\pm$ 0.028	0.838 $\pm$ 0.029	0.682 $\pm$ 0.034	0.857 $\pm$ 0.026	0.854 $\pm$ 0.016
precision macro	0.613 $\pm$ 0.054	0.727 $\pm$ 0.111	0.676 $\pm$ 0.046	0.758 $\pm$ 0.084	0.693 $\pm$ 0.054
recall macro	0.608 $\pm$ 0.052	0.587 $\pm$ 0.051	0.702 $\pm$ 0.044	0.615 $\pm$ 0.059	0.581 $\pm$ 0.029
f1 macro	0.607 $\pm$ 0.050	0.619 $\pm$ 0.060	0.633 $\pm$ 0.033	0.647 $\pm$ 0.066	0.599 $\pm$ 0.033
precision weighted	0.783 $\pm$ 0.028	0.816 $\pm$ 0.045	0.818 $\pm$ 0.027	0.833 $\pm$ 0.035	0.804 $\pm$ 0.021
recall weighted	0.783 $\pm$ 0.028	0.838 $\pm$ 0.029	0.682 $\pm$ 0.034	0.857 $\pm$ 0.026	0.854 $\pm$ 0.016
f1 weighted	0.782 $\pm$ 0.027	0.812 $\pm$ 0.031	0.715 $\pm$ 0.029	0.829 $\pm$ 0.030	0.812 $\pm$ 0.018

DT: Decision Tree; KNN: k -Nearest Neighbors; NB: Naïve Bayes; RF: Random Forest; SVM: Support Vector Machine.

How we selected the 5 features?

Metaheuristic??

- 1)Total hardness
- 2)Orthophosphate
- 3)PH
- 4)Ammonia
- 5)Total suspended solids



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Drinking Water Resources for the General Public

Water Testing Assistance Program

Conclusion

The final idea

Assess water.
Cost: Almost nothing